

Environmental Cognition-Driven Hierarchical Decision-Making for Air-Ground Collaborative Communication

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Abstract—The unparalleled communication and sensing capabilities of the 6th generation (6G) networks can fulfill the diverse requirements of emerging scenarios like the metaverse. However, several issues such as queue transmission between numerous heterogeneous nodes and difficulties in understanding complex and dynamic environments from high-level cognition urgently need to be addressed. In response, this paper proposes an environmental cognition-driven method based on graph convolutional networks (GCNs), which comprehensively considers node positions, task queues, and frequency conflicts, then optimizing node clustering to cope with complex and dynamic electromagnetic environments. In the meanwhile, to overcome the limitations of conventional reinforcement learning (RL) methods in addressing the increasing complexities of multi-task and multi-granularity decision-making, a hierarchical reinforcement learning (HRL) method based on the actor-critic (AC) framework is proposed to quick decision-making for air-ground collaborative communication. Numerical results demonstrate that the proposed environmental cognition-driven hierarchical decision-making method can reduce redundant interactions among nodes and improve the transmission rate.

Index Terms—environmental cognition-driven, air-ground collaborative communication, hierarchical reinforcement learning, graph convolutional network, multi-agent spectrum allocation.

I. INTRODUCTION

The deployment of 6G networks encounters numerous challenges, particularly in the distribution and allocation of spectrum resources. The growing demand for high-throughput applications [1] and massive device access [2] has made it crucial to address congestion effectively and establish intelligent connections between users and devices. However, in complex urban settings, scattered buildings construct intricate electromagnetic environments [3]. These buildings affect the transmission of communication signals and also serve as barriers against malicious interference. Currently, there exists a deficiency in efficient air-ground collaborative communication methods that comprehensively exploit the spatial and frequency domain resources in complex urban environments from high-level cognition to effectively counter dynamic interference.

Air-ground collaborative communication utilizes the maneuverability of unmanned aerial vehicles (UAVs) to provide dynamic network deployment and assist in unloading task

queues. To address the dynamic computational demands of air-ground collaborative communication, ref. [4] proposed a joint online trajectory planning and offloading scheduling method based on RL. By formulating trajectory planning and offloading scheduling as a coupled Markov decision process (MDP), this method enables adaptive decision-making that accommodate spatiotemporal changes effectively. Under the condition of shared spectrum resources, ref. [5] addressed the trade-off issue of UAV-integrated sensing and communication systems by jointly optimizing UAV trajectory, transmitting beam shaping, and sensing task execution time under the constraints of sensing frequency and beam pattern gain, maximizing the communication rates of users. It is worth noting that the aforementioned studies neglect the environment's effect on agent decision-making and collaborative decision-making among multiple agents. The agents can cognize changes in the surrounding environment, optimizing the utilization of communication resources and improving communication efficiency. In large-scale multi-agent RL scenarios, agents do not necessarily need to observe the states or actions of all participants, focusing on relevant agents in the vicinity suffices [6]. Hence, clustering agents with similar tasks in complex urban environments reduces storage and computational overhead, improving the decision-making efficiency.

Conventional RL methods are hindered in achieving fast convergence and efficient communication performance in 6G networks due to their sensitivity to hyperparameters, reliance on a single strategy [7], and limitations in solving multidimensional decision-making problems. In terms of solving such decision-making problems, ref. [8] decomposed the scheduling problem into two sub-problems, utilizing a HRL method to efficiently schedule UAV computing tasks, and enhancing solution efficiency through hierarchical alternating optimization. Ref. [9] decomposed the complex decision-making problem of joint optimization of UAV trajectories, task association, computation, and transmission resource allocation into three sub-problems, then proposing a heuristic hybrid scheduling algorithm based on RL to minimize the energy consumption of user devices. Briefly, HRL can decompose complex decision-making tasks into sub-tasks at different layers, thereby reducing the complexity of the state and action space, mitigating the curse of dimensionality, and improving the overall performance and learning efficiency.

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In this paper, we investigate the air-ground collaborative communication issues for 6G networks in complex urban environments, as shown in Fig. 1. Basically, this paper focuses on reducing redundant interactions between nodes, so as to improve the overall transmission efficiency of the system. Nevertheless, solving the queue accumulation issue for air-ground collaborative communication in complex urban environments is nontrivial. Therefore, we firstly propose a novel environmental cognition-driven decision-making method, which understands complex and dynamic environments from high-level cognition and utilizes GCN to aggregate node features for clustering. In order to model the real complex urban environment, we consider small-scale fading and signal attenuation caused by buildings occlusion, and use the logarithmic distance decay model to describe the air-ground channel. Moreover, employing HRL based on the AC framework is beneficial for addressing multi-task and multi-granularity decision-making in 6G networks. This method can avoid the curse of dimensionality, improving overall performance and achieving faster convergence. Simulation results show that the proposed environmental cognition-driven hierarchical decision-making algorithm based on AC framework optimizes node clustering through environmental information, reduces redundant interactions between nodes, thus improving queue transmission rate.

The remainder of this paper is organized as follows. Section II elaborates on the system model. Section III formulates the design problem. Section IV proposes a environmental cognition-driven hierarchical decision-making algorithm. Section V discusses the simulation results. Conclusions are drawn in Section VI.

II. SYSTEM MODEL

As shown in Fig. 1, this paper focuses on the scenario of complex urban environments, where UAVs and ground nodes utilize urban buildings and spectrum resources to counter dynamic interference. In this scenario, rather than using conventional cellular networks with base stations, a cell-free network connects M UAVs and N ground nodes through numerous small antennas distributed across the entire area [10]. The set of UAVs can be represented as $U = \{U_1, U_2, \dots, U_m\}, m = 1, 2, \dots, M$, and the set of ground nodes can be represented as $u = \{u_1, u_2, \dots, u_n\}, n = 1, 2, \dots, N$. The ground nodes are distributed in K different clusters based on environmental characteristics. Moreover, maintain the UAVs at a constant altitude H while ensuring that ground nodes and jammers are in a fixed position on the horizontal plane.

A. Communication Channel Model

This paper describes the air-ground channel using a logarithmic distance decay model, which considers small-scale fading between buildings and signal attenuation caused by buildings occlusion in complex urban environments [11]. The transmission rate of the ground node n can be formulated as $R_n(t) = \log_2(1 + P \cdot \gamma_n / N)$, where P and N respectively represent the transmission power and noise power of ground node

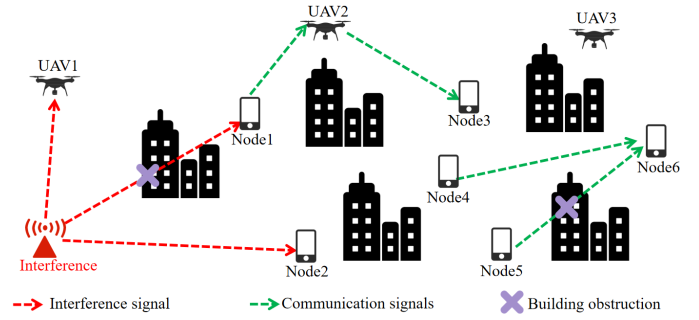


Fig. 1. Air-ground collaborative communication in the urban scenario.

n , and γ_n represents the path loss. The air-ground transmission path loss can be expressed as $\gamma_n = d_n(t)^{-\alpha} \cdot 10^{X_{Rayleigh}/10} \cdot \beta$, where α represents the path loss index of vacuum; $X_{Rayleigh}$ represents that small-scale fading can be characterized as a Rayleigh-distributed random variable with a scaling factor of $\sigma = 1$, and the signal attenuation induced by buildings can be modeled using a discrete factor β , $\beta \in \{1, 0.01\}$. When the signal transmission is obstructed, $\beta = 0.01$; otherwise, $\beta = 1$. The path loss of ground-to-ground transmission between ground nodes can be calculated as

$$P_r = P_t \frac{G_r G_t}{L} \left(\frac{\lambda}{4\pi} \right)^2 \left(\frac{1}{d} \right)^\gamma, \quad (1)$$

where P_r represents the received signal power; P_t represents the transmitted signal power; G_r represents the receiving antenna gain; G_t represents the transmitting antenna gain; L represents system losses; λ represents the signal wavelength; d represents the distance between the transmitter and receiver; γ represents the path loss exponent.

B. Communication Interference Model

In the scenario of complex urban environment, there are three communication methods for ground nodes to address randomly generated task queues in each timeslot, namely no communication, direct communication, and communication through UAVs. Moreover, task queues generated by a single ground node in a single timeslot can only be transmitted to a single UAV. As shown in Fig. 1, the jammers are maliciously disrupting the queue transmission between the ground nodes. The jammer's interference signal operates on a frequency F_j , $F_j = \{f_1, f_2, \dots, f_j\}, j \in \{1, 2, \dots, J\}$ that repeats in accordance with the timeslot cycle. When a ground node selects frequency band F , $F = \{f_1, f_2, \dots, f_i\}, i \in \{1, 2, \dots, I\}$ for communication, and its selected frequency aligns with the interference signal transmitted by a jammer in the current timeslot, interference is deemed to occur. If multiple nodes choose the same frequency point for communication, it can be considered that there is mutual interference between nodes. When mutual interference occur, the transmission rate can be expressed as

$$R_n(t) = \log_2 \left(1 + \frac{P \cdot \gamma_n}{G_j P_j + \sum_{a \in A} G_i P + N} \right), \quad (2)$$

where G_j represents the channel gain of the interference signal at the interfered nodes; P_j represents the interference signal

power; A represents the set of nodes that interfere with each other; G_i represents the channel gain of the interference signal.

C. Queue Transmission Model

Assume that the random queue generated by each ground node in single timeslot is Δ_t , the accumulated queue for each ground node is Δ_n , and the accumulated queue for each UAV is Δ_m . It can be also assumed that each timeslot can be divided into half, that is $\tau/2$ timeslot for transmission and the other half for reception. Each ground node transmits its queue to another ground node during each timeslot, and the receiving ground node only receives the queue from the corresponding transmitting ground node. At the beginning of a timeslot, the ground node generates random queue, i.e. $\Delta_n = \Delta_n + \Delta_t$. When a ground node chooses to communicate through UAVs, if the transmission can be completed, i.e. $\Delta_t \leq R_n(t) \cdot \tau/2$; in the uplink, the UAV receives the current queue, and the ground node's queue is reduced, that is $\Delta_m = \Delta_m + \Delta_t$ and $\Delta_n = \Delta_n - \Delta_t$; in the downlink, the UAV forwards the current queue, i.e. $\Delta_m = \Delta_m - \Delta_t$. If the transmission cannot be completed, i.e. $\Delta_t > R_n(t) \cdot \tau/2$; in the uplink, the UAV receives the maximum amount of queue it can receive, and the ground node's queue is increased by the maximum amount of queue it can transmit, that is $\Delta_m = \Delta_m + R_n^{uplink}(t) \cdot \tau/2$ and $\Delta_n = \Delta_n - R_n^{uplink}(t) \cdot \tau/2$; in the downlink, the UAV forwards the maximum amount of queue it can forward, i.e. $\Delta_m = \Delta_m - R_n^{downlink}(t) \cdot \tau/2$, and the unsent queue is transmitted in the following timeslot. When the ground node chooses direct communication, if the transmission can be completed, i.e. $\Delta_t \leq R_n(t) \cdot \tau/2$, subtract the current queue from the queue of ground nodes, i.e. $\Delta_n = \Delta_n - \Delta_t$; if the transmission cannot be completed, i.e. $\Delta_t > R_n(t) \cdot \tau/2$, subtract the maximum amount of queue it can forward, that is $\Delta_n = \Delta_n - R_n(t) \cdot \tau/2$, and the unsent queue is transmitted in the following timeslot.

III. PROBLEM FORMULATION

Due to the continuous queue generation at each node throughout the timeslot, communication decision-making made within this timeslot cannot consider the current amount of queue transmitted. Therefore, the joint strategy for air-ground collaborative communication and UAV trajectory planning involves a sequential decision-making problem, which can be boiled down to a MDP. However, simply modeling multiple decision problems as MDPs cannot solve the problem of ground node communication and cluster association decision-making requiring crossing multiple timeslots, nor can it solve the sparse reward problem of cluster association. Moreover, this will undoubtedly lead to a significant increase in the complexity of the state space and action space of the agent, resulting in a curse of dimensionality and a waste of spectrum resources.

When addressing multi-task and multi-granularity decision-making issues with different timeslots, we employ a semi-Markov decision process [12] to model this issue and achieve joint decision-making and policy learning at different timeslots. This paper divides the air-ground collaborative communication scenario into three layers: the ground node communication

decision-making layer, the UAV trajectory planning layer and the cluster association decision-making layer. The state of the ground node can be represented by its observations, that is $s_{n,t} = [p_m, p_n, \mathbf{a}_{n,t-1}, f_n, q_n]$, where p_m represents the position of the selected UAV; $\mathbf{a}_{n,t-1}$ represents the action from the previous timeslot; p_n, f_n, q_n respectively represent the position, the occupied frequency point and the accumulated queue of the ground node. Then, the top layer sends the decision of ground nodes $\mathbf{a}_{n,t}$ to the middle layer, and this decision can be shared between UAVs. In the middle layer, the UAV positions, ground node decision-making, queue accumulation of ground nodes, and spectral information are utilized for UAV trajectory planning, denoted as $\mathbf{a}_{m,t}$. In timeslot t , the state of UAV m includes UAV position, queue accumulation, and cluster decision actions in the previous timeslot, i.e. $s_{m,t} = [p_m, q_m, \mathbf{a}_{k,t-1}]$. Finally, the bottom layer receives the ground node clustering and decision-making issued by the top layer, as well as the UAV positions issued by the middle layer. Based on this, collaborative decision-making between the ground node clusters and the UAVs can be carried out, that is $\mathbf{a}_{k,t}$. In timeslot t , the state of cluster association decision-making includes the positions of all UAVs and the cluster association decision-making in the previous timeslot, i.e. $s_{k,t} = [p_1, \dots, p_M, \mathbf{a}_{k,t-1}]$.

In timeslot t , the ground node n in the top layer receives a reward $r_{k,n,t}$, which represents the accumulated value of the current timeslot queue of the ground node, i.e. $r_{k,n,t} = q_n$. After Δ timeslots, the reward obtained by node cluster k in the bottom layer can be defined as the accumulation of queue between all ground nodes and UAVs, i.e. $r_{k,t+\Delta} = \sum_{\Delta} (\sum_{m=1}^M q_m + \sum_{n=1}^N q_n)$. After Δ timeslots, the reward obtained by UAV m in the middle layer can be defined as the accumulation of queue between the UAV and associated ground nodes, i.e. $r_{m,t+\Delta} = \sum_{\Delta} q_m + \sum_{n=1}^{N/M} q_n$.

Aiming at avoiding frequency interference and enhancing the throughput of air-ground collaborative communication in the complex and dynamic electromagnetic environment, the optimization function for collaborative decision-making between ground node communication and UAV trajectory planning can be defined as minimizing the accumulated queue of all ground nodes. The optimization function can be cast as

$$\min_{\pi_n} E \left(\sum_{t=1}^{\infty} \sum_{n=1}^N \lambda^{t-1} q_n \right), \quad (3)$$

where $\pi_n : \mathbf{S} \rightarrow \mathbf{A}$ represents a deterministic policy w.r.t. the communication selection at node n ; $\lambda \in (0, 1)$ represents a discount factor.

IV. ENVIRONMENTAL COGNITION-DRIVEN HIERARCHICAL DECISION-MAKING ALGORITHM

According to Section III, in certain multi-agent tasks, excessive interactions among agents can lead to redundant information, thereby compromising decision-making. This issue becomes more pronounced as the number of agents increases, complicating interactions and exacerbating the impact of redundant information. In response, we propose an environmental

cognition-driven method aimed at reducing these redundancies by cancelling the pre-fixed node clusters and introducing GCN [13] for environmental cognition. This method comprehensively considers buildings occlusion, node positions, task queues, and frequency relationships in the complex urban environment.

In the ground node communication decision-making layer, all ground nodes are abstracted into a graph representation as $\mathbf{G} = (\mathbf{V}, \mathbf{E})$, where $\mathbf{V}$ represents the set of nodes and $\mathbf{E}$ represents the edge set. Firstly, construct the adjacency matrix $\mathbf{A}$ and the feature matrix $\mathbf{X}$. The adjacency matrix $\mathbf{A}$ has a dimension of $N \times N$, which reflects the communication relationship between nodes. If there is communication between node i and node j , $i, j \in \{1, 2, \dots, N\}$, then $\mathbf{A}_{ij} = 1$; otherwise, $\mathbf{A}_{ij} = 0$. The feature matrix $\mathbf{X}$ has a dimension of $N \times F$, where F represents the number of node features. Secondly, construct a GCN model, where the number of neurons in the input layer can be equal to the number of columns in the feature matrix $\mathbf{X}$, and the number of neurons in the output layer can be determined by the number of clusters. By using the Glorot method [14] that follows a uniform distribution, the weight matrix $\mathbf{W}$ can be initialized as $\mathbf{W} \sim U(-\sqrt{6}/\sqrt{n_j + n_{j+1}}, \sqrt{6}/\sqrt{n_j + n_{j+1}})$, where n_j and n_{j+1} represent the number of neurons in the input layer and the output layer, respectively. Then, the adjacency matrix $\mathbf{A}$ and the feature matrix $\mathbf{X}$ can be separately preprocessed by $\hat{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{A} + \mathbf{I})\mathbf{D}^{1/2}$ and $\hat{\mathbf{X}} = (\mathbf{X} - \bar{\mathbf{X}})/\delta_{\mathbf{X}}$, where $\mathbf{D}$ represents the degree matrix of the adjacency matrix $\mathbf{A}$; $\mathbf{I}$ represents the identity matrix; $\bar{\mathbf{X}}$ and $\delta_{\mathbf{X}}$ represent the mean and the variance, respectively. Finally, the adjacency matrix $\mathbf{A}$ and feature matrix $\mathbf{X}$ are input into the GCN, and the prediction result $\mathbf{Y}$ can be obtained through forward propagation, that is

$$\mathbf{Y} = \text{softmax}(\hat{\mathbf{A}} \text{ReLU}(\hat{\mathbf{A}}\hat{\mathbf{X}}\mathbf{W}^{(0)})\mathbf{W}^{(1)}), \quad (4)$$

where $\hat{\mathbf{A}}$ and $\hat{\mathbf{X}}$ represent the preprocessing of the adjacency matrix $\mathbf{A}$ and the feature matrix $\mathbf{X}$, respectively.

In GCN, the calculation process from the hidden layer l to the hidden layer $l + 1$ can be expressed as

$$\mathbf{H}^{l+1} = \sigma(\hat{\mathbf{A}}\mathbf{H}^l\mathbf{W}^l), \quad (5)$$

where σ represents the ReLU activation function. During the training process, we use a cross-entropy loss function and update the weight matrix $\mathbf{W}$ through backpropagation. The cross-entropy loss function can be calculated as

$$\text{Loss} = -\sum_{i=1}^N \sum_{j=1}^K \mathbf{Y}_{ij} \log(\mathbf{P}_{ij}), \quad (6)$$

where N represents the number of nodes and K represents the number of clusters; $\mathbf{P}_{ij}$ and $\mathbf{Y}_{ij}$ represent the predicted value and label value, respectively.

Similarly, in the cluster association decision-making layer, all ground nodes and UAVs are abstracted as graph representations. Construct an adjacency matrix $\mathbf{A}$, reflecting the queue transmission relationship between clusters. Then, construct a feature matrix $\mathbf{X}$ based on node positions and transmission queue. Finally, construct a new GCN model and iteratively train it to obtain clustering results. In the UAV trajectory planning

layer, UAVs' decision-making do not involve clustering and do not apply the environmental cognition-driven method.

In the AC algorithm, the actor consists of a policy network $\pi(s; \theta)$ that generates the agent's actions $\mathbf{a}_t$ based on observed environment $\mathbf{s}_t$ using parameterization strategies. The critic, on the other hand, evaluates these actions using a parameterized state value function $V(s; w)$ or action value function $Q(s, a; w)$ and uses temporal-difference (TD) error to guide the strategy network and value network parameters update. In this paper, we use two fully connected neural networks to approximate the policy function $\pi(s; \theta)$ and the state value function $V(s; w)$. Both networks take agent observations $\mathbf{s}_t$ denoted as inputs. The value network output $V(\mathbf{s}_t)$ corresponds to the state $\mathbf{s}_t$, while the policy network outputs the probabilities of actions for the given state $\mathbf{s}_t$. Moreover, the ground node communication decision-making layer, the UAV trajectory planning layer, and the cluster association decision-making layer have discrete action spaces, leading to the adoption of a softmax strategy.

Define the policy network θ_k and value network w_k for ground node cluster k . The decision-making process for cluster association at timeslot t can be denoted as $\mathbf{a}_{k,t} = \pi(\cdot | \mathbf{s}_{k,t}; \theta_k)$, where $\pi(\cdot | \mathbf{s}_{k,t}; \theta_k)$ represents the selection of $\mathbf{a}_{k,t}$ with certain probability among the various options outputted by parameterized policy network θ_k and satisfies $\sum_{\mathbf{a}_{k,t} \in A} \pi(\mathbf{s}_{k,t}, \mathbf{a}_{k,t}; \theta_k) = 1$. After Δ timeslots, when this scheme concludes, the TD error can be computed based on the cumulative reward $r_{k,t+\Delta}$ and the output of the value network, that is

$$\delta_{k,t} = r_{k,t+\Delta} + \gamma V(\mathbf{s}_{k,t+\Delta}; w_k) - V(\mathbf{s}_{k,t}; w_k). \quad (7)$$

The value network updates its parameters with the objective of minimizing the mean square error of the TD error. The value network parameters can be updated by

$$w_k = w_k - \alpha \cdot \nabla L(w_k), \quad (8)$$

where $L(w_k)$ represents the loss function, i.e.

$$L(w_k) = [r_{k,t+\Delta} + \gamma V(\mathbf{s}_{k,t+\Delta}; w_k) - V(\mathbf{s}_{k,t}; w_k)]^2. \quad (9)$$

The policy network updates its parameters along the gradient direction of the loss function $J(\theta)$ guided by TD error. The loss function can be obtained as

$$J(\theta_k) = [\log \pi(\mathbf{s}_{k,t}; \theta_k) \cdot (r_{k,t+\Delta} + \gamma V(\mathbf{s}_{k,t+\Delta}; w_k) - V(\mathbf{s}_{k,t}; w_k))], \quad (10)$$

and the policy network parameters can be updated by

$$\theta_k = \theta_k - \alpha \cdot \nabla J(\theta_k). \quad (11)$$

Similarly, define the policy network θ_m and value network w_m for UAV m , the UAV trajectory planning decision at timeslot t can be defined as $\mathbf{a}_{m,t} = \pi(\cdot | \mathbf{s}_{m,t}; \theta_m)$. When the scheme concludes after Δ timeslots, calculate the TD error for UAV m based on the cumulative reward $r_{m,t+\Delta}$ and the output of the value network, that is $\delta_{m,t} = r_{m,t+\Delta} + \gamma V(\mathbf{s}_{m,t+\Delta}; w_m) - V(\mathbf{s}_{m,t}; w_m)$, and the updating methods for the value network w_m and policy network θ_m are the same as equation (8) and (11).

Algorithm 1 HRL with Environmental Cognition-Driven

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1: Initialize: Initialize  $\mathbf{A}$ ,  $\mathbf{X}$ , and parameters of GCN;  $\mathbf{s}_{k,0}$ ,  $\mathbf{a}_{k,0}$ ,  $r_{k,0}$ ,  $\theta_k$  and  $w_k$  of ground node cluster  $k$ ;  $\mathbf{s}_{m,0}$ ,  $\mathbf{a}_{m,0}$ ,  $r_{m,0}$ ,  $\theta_m$  and  $w_m$  of UAV  $m$ ;  $\mathbf{s}_{k,n,0}$ ,  $\mathbf{a}_{k,n,0}$ ,  $r_{k,n,0}$ ,  $\theta_{k,n}$  and  $w_{k,n}$  of ground node  $n$ .
2: repeat
3:   for  $T$  to  $T_{max}$  do
4:     In timeslot  $T$ , creat  $\mathbf{A}$  and  $\mathbf{X}$ , and input them into the pre-trained GCN to obtain ground node clusters.
5:     Ground node cluster  $k$  obtains  $\mathbf{s}_{k,T}$ , and generates  $\mathbf{a}_{k,T} = \pi(\cdot|\mathbf{s}_{k,T}; \theta_k)$  based on policy network.
6:     UAV  $m$  obtains  $\mathbf{s}_{m,T}$ , and generates  $\mathbf{a}_{m,T} = \pi(\cdot|\mathbf{s}_{m,T}; \theta_m)$  based on policy network.
7:     Creat  $\mathbf{A}$  and  $\mathbf{X}$  based on  $\mathbf{s}_{k,T}$  and  $\mathbf{s}_{m,T}$ , and feed them into pre-trained GCN to obtain cluster association results, which are then integrated into  $\mathbf{s}_{k,T+1}$ .
8:     for  $t$  to  $t_{max}$  do
9:       Ground node  $n$  obtains  $\mathbf{s}_{k,n,t}$  and generates  $\mathbf{a}_{k,n,t} = \pi(\cdot|\mathbf{s}_{k,n,t}; \theta_{k,n})$  based on policy network.
10:      At the end of timeslot  $t$ , ground node  $n$  receives  $r_{k,n,t}$  and updates  $\theta_{k,n}$  and  $w_{k,n}$ .
11:    end for
12:    At the end of timeslot  $T+t$ , ground node cluster  $k$  receives  $r_{k,T+t}$  and updates  $\theta_k$  and  $w_k$ .
13:    At the end of timeslot  $T+t$ , UAV  $m$  receives  $r_{m,T+t}$  and updates  $\theta_m$  and  $w_m$ .
14:  end for
15: until Stopping criteria

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Unlike the two aforementioned schemes, the communication decision action of the ground node needs to be made within each timeslot. Define the policy network $\theta_{k,n}$ and value network $w_{k,n}$ for ground node n within cluster k , which have the same updating methods as equation (8) and (11). The communication decision action of the ground node can be defined as $\mathbf{a}_{k,n,t} = \pi(\cdot|\mathbf{s}_{k,n,t}; \theta_{k,n})$, and the TD error can be calculated as $\delta_{k,n,t} = r_{k,n,t} + \gamma V(\mathbf{s}_{k,n,t+1}; w_{k,n}) - V(\mathbf{s}_{k,n,t}; w_{k,n})$.

The pseudo code of the algorithm is shown in Algorithm 1.

V. SIMULATION RESULTS

In the simulations, we established three clusters of ground nodes, each consisting of two ground nodes, and deployed three UAVs. The UAVs maintain a constant altitude of 50m and each UAV moves 10m per timeslot. Initially, UAV 1 is positioned at (50, 100, 50), UAV 2 at (100, 100, 50), and UAV 3 at (150, 100, 50); the ground node positions are initialized as (10, 50, 0), (130, 100, 0), (100, 60, 0), (180, 180, 0), (50, 80, 0), and (100, 150, 0); the jammer is located at (0, 0, 0). The node has six frequency bands to choose from, with a bandwidth of $B=15\text{MHz}$. Moreover, ground nodes generate queues randomly in each timeslot and determine whether to transmit and with which node. When transmission is needed, the queue sent by each ground node in each timeslot is generated as a sawtooth wave. Power configurations ground nodes transmitting at 10W, juxtaposed with a jammer operating at 10000W, with noise power set at 10^{-12}W . The loss index of empty ground path is represented by $\gamma = 2$, and the loss index of ground ground path is $\gamma = 6$. The simulation experiment was conducted using MATLAB R2021a.

As shown in Fig. 2, the blue curve reflects the accumulated queues of UAVs. It is evident that initially, each UAV experi-

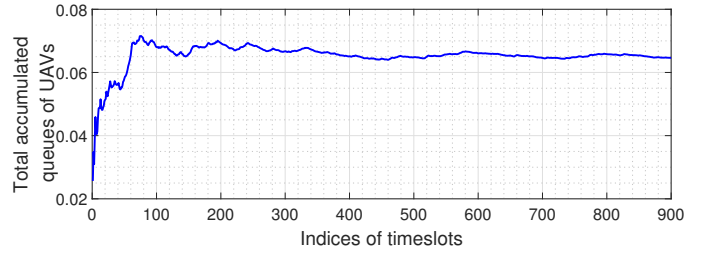


Fig. 2. Total accumulated queues of UAVs.

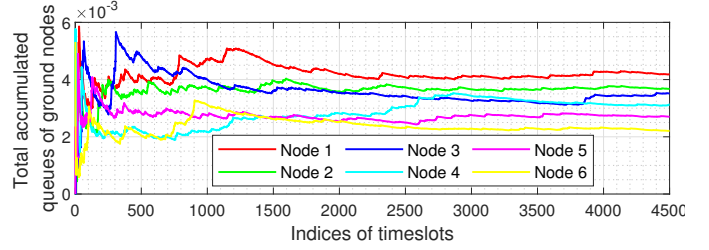


Fig. 3. Total accumulated queues of ground nodes.

ences a lower transmission rate per timeslot. As the algorithm refines node decisions, each node enhances its decision-making, progressively elevating the overall system transmission rate until convergence. The different colored curves in Fig. 3 illustrate the accumulated queues of each ground node in the scenario. Similarly, through hierarchical decision-making and multiple iterations, the decision-making of ground nodes can converge and accelerate the overall transmission rate.

Incorporating a GCN into the ground node communication decision-making layer, where the adjacency matrix $\mathbf{A}$ reflects the queue transmission relationship between ground nodes, and the feature matrix $\mathbf{X}$ includes node positions and whether frequency interference occurs between ground nodes. The simulation results are shown in Fig. 4, where the blue dashed curve shows the loss function between predicted values and label values, and the red solid curve shows the accuracy of predicted values. With the convergence in the training set and the improvement of prediction accuracy, it can be believed that the offline trained GCN can be applied to ground node communication decision-making. Similarly, incorporating a GCN into the cluster association decision-making layer, where the adjacency matrix $\mathbf{A}$ reflects the queue transmission relationship between ground node clusters and UAVs, and the feature matrix $\mathbf{X}$ includes node positions and task queues. Based on the simulation results in Fig. 5, it can be believed that the offline trained GCN can be applied to the association decision-making between ground node clusters and UAVs.

As shown in Fig. 2, as the algorithm optimizes node decision-making, the agents make better decision-making, resulting in an increasing trend in transmission rate until convergence. In the simulations, we calculate the total accumulated queues of all UAVs to compare the performance of methods with and without environmental cognition. As shown in Fig. 6(a), the green curve represents the accumulated queues of UAVs without environmental cognition, while the red curve represents the accumulated queues of UAVs with environmental cognition in the ground node communication decision-making layer. With

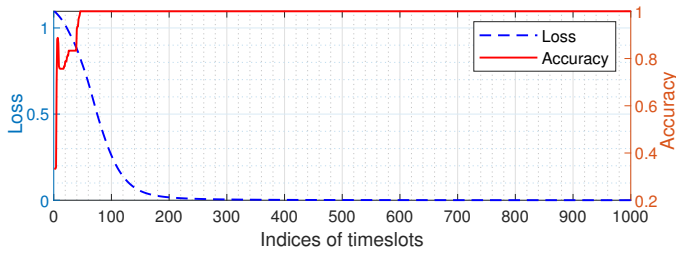


Fig. 4. The loss and accuracy of GCN in the ground node communication decision-making layer.

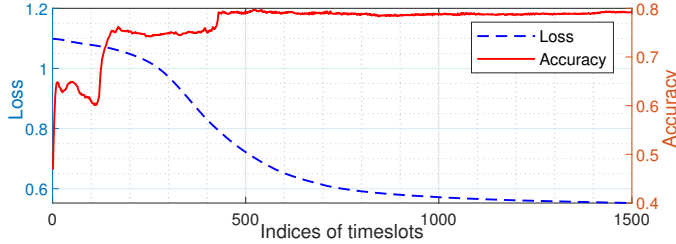
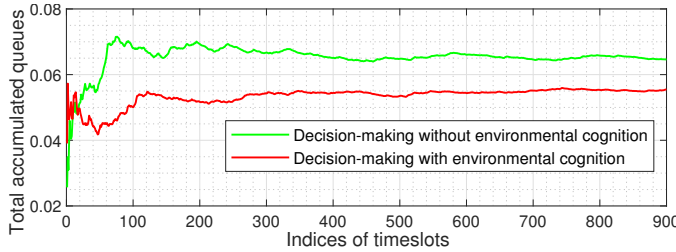
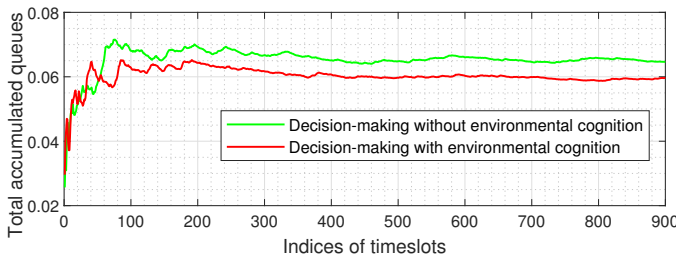


Fig. 5. The loss and accuracy of GCN in the cluster association decision-making layer.

the iterative training of the algorithm, node decision-making converges. Similarly, in Fig. 6(b), the green curve represents the accumulated queues of UAVs without environmental cognition, while the red curve represents the accumulated queues of UAVs with environmental cognition in the cluster association decision-making layer. From the comparison of simulation results, it can be seen that after incorporating environmental cognition-driven scheme, the accumulated queue of UAVs is less, and the total transmission rate of the system is better. The above simulation results demonstrate that in complex urban environments, the environmental cognition-driven multi-agent hierarchical decision-making algorithm can better improve the overall queue transmission rate of nodes by exploring high-level cognition of the surrounding environment.



(a) In the ground node communication decision-making layer.



(b) In the cluster association decision-making layer.

Fig. 6. Comparison of total accumulated queues with and without environmental cognition.

VI. CONCLUSIONS

This paper has investigated air-ground collaboration communication issues for 6G networks in complex urban environments. To this end, we have proposed a novel environmental cognition-driven hierarchical decision-making method, which utilizes GCN to comprehensively consider node positions, task queues, and frequency conflicts. Numerical results illustrate that by integrating the environmental cognition-driven scheme, the proposed hierarchical decision-making algorithm based on the AC framework can optimize node clustering to reduce redundant interactions between nodes. Benefiting from this, the proposed method can understand complex and dynamic environments from high-level cognition, thereby improving the transmission rate between ground nodes.

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